

Exercise 5 - Magnetic exchange coupling in a Cu(II) dimer from broken-symmetry DFT

The compound is a paddle-wheel dimer: four trifluoroacetate bridges between the two Cu(II) ions (Cu–Cu 2.73 Å) and one alkoxide-type O donor on each axial position; total charge –2 as given. Each Cu(II) is d^9 with $S = 1/2$. Keywords: ! UKS B3LYP <basis> <auxiliary basis> RIJCOSX TightSCF SlowConv with %scf BrokenSym 1,1 end and the coordinate line * xyzfile –2 3, i.e. the high-spin multiplicity. ORCA first converges the high-spin state (HS, both spins parallel), then flips the spin on one Cu and converges the broken-symmetry state (BS). The three runs differ only in the relativistic treatment: non-relativistic (def2-SVP, def2/J), DKH (DKH-def2-SVP, SARC/J) and ZORA (ZORA-def2-SVP, SARC/J); the functional and the basis-set size are the same in all three, so the table isolates the relativistic effect. The assignment does not prescribe the non-relativistic level, so B3LYP/def2-SVP was chosen to match the other two.

Results

Table 1. Total energies of the high-spin and broken-symmetry states, their spin expectation values and the exchange coupling constants in cm^{-1} . ORCA uses $H = -2J S_A \cdot S_B$, so $J < 0$ means antiferromagnetic coupling. $J(1)$ is the Ginsberg–Noodleman expression, $J(2)$ the Bencini–Gatteschi one and $J(3)$ the Yamaguchi expression $J = (E_{BS} - E_{HS}) / (\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS})$.

The last three rows repeat the calculations with a finer integration grid (DefGrid3) and VeryTightSCF.

Run	E(HS) / E_h	E(BS) / E_h	$\langle S^2 \rangle_{HS}$	$\langle S^2 \rangle_{BS}$	E(HS)–E(BS) / cm^{-1}	J(1)	J(2)	J(3)
non-relativistic	–6076.975267	–6076.975094	2.0050	1.0006	–38.0	+38.0	+19.0	+37.8
DKH	–6107.249506	–6107.249605	2.0041	0.9989	+21.7	–21.7	–10.9	–21.6
ZORA	–6122.764932	–6122.765176	2.0038	0.9976	+53.5	–53.5	–26.8	–53.2
non-relativistic, tight grid	–6076.975278	–6076.975105	2.0050	1.0006	–37.9	+37.9	+19.0	+37.8
DKH, tight grid	–6107.249520	–6107.249620	2.0041	0.9989	+21.9	–21.9	–10.9	–21.8
ZORA, tight grid	–6122.765126	–6122.765371	2.0038	0.9976	+53.7	–53.7	–26.9	–53.4

The broken-symmetry states are the intended ones: the Mulliken spin populations on the two Cu atoms are +0.63 and –0.63 (non-relativistic; +0.72 / –0.72 with ZORA), and $\langle S^2 \rangle_{BS} \approx 1.0$ against $\langle S^2 \rangle_{HS} \approx 2.0$, which is what two localised, antiparallel $S = 1/2$ centres give.

What the table shows

The relativistic correction to the total energy is enormous, and irrelevant for J. DKH lowers the total energy by 30.3 E_h and ZORA by 45.8 E_h (6.6 and 10.0 million cm^{-1}). This is almost entirely the contraction and stabilisation of the Cu 1s–2p core shells, which is the same in both spin states and therefore cancels in $E(HS) - E(BS)$. The two relativistic Hamiltonians are also not on the same absolute scale, so only differences within one Hamiltonian are meaningful.

J itself comes from a tiny energy difference. $E(HS) - E(BS)$ is only 22–54 cm^{-1} , i.e. about $244 \times 10^{-6} E_h$: 7 orders of magnitude smaller than the total energy and 5 orders smaller than the relativistic correction.

The three formulas differ by factors, not by physics. $J(1)$ and $J(3)$ agree closely because $\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS} \approx 1$ here, which is the weak-coupling limit that the Yamaguchi expression interpolates to; $J(2)$ is

smaller by a factor $S_{\max}(S_{\max}+1)/S_{\max}^2 = 2$ by construction. Which one is quoted matters as much as the relativistic treatment, so the convention has to be stated.

Relativistic effects on J with the prescribed SVP basis. $J(3)$ goes from $+37.8 \text{ cm}^{-1}$ (non-relativistic) to -21.6 cm^{-1} (DKH) and -53.2 cm^{-1} (ZORA). The changes, -59 and -91 cm^{-1} , are tiny compared with the change of the total energy, but as large as J itself, and at this level they even change its sign (weakly ferromagnetic without, weakly antiferromagnetic with relativity). The two relativistic Hamiltonians also disagree with each other by 32 cm^{-1} . Because such small energy differences are easily distorted, the SVP results were checked for numerical and basis-set convergence below.

Numerical check. $E(\text{HS}) - E(\text{BS})$ is of the same order as the error of the chain-of-spheres approximation for the exchange integrals, so all three calculations were repeated with a finer integration grid (DefGrid3) and VeryTightSCF (last three rows of Table 1). $J(3)$ changes by only -0.07 , -0.13 , -0.15 cm^{-1} , less than 0.1 cm^{-1} : the numbers are converged with respect to the numerical integration.

Are the SVP results converged? Basis set and functional

Table 2. Additional B3LYP calculations with the larger TZVP-level basis sets (upper block, compared with the SVP results of Table 1) and non-relativistic SVP calculations with functionals containing 10%, 20% and 25% exact exchange (lower block). Same geometry, charge and BrokenSym setup throughout.

Run	$E(\text{HS})-E(\text{BS}) / \text{cm}^{-1}$	$J(3) / \text{cm}^{-1}$	Coupling
B3LYP/def2-SVP, non-relativistic	-38.0	+37.8	ferromagnetic
B3LYP/DKH-def2-SVP, DKH	+21.7	-21.6	antiferromagnetic
B3LYP/ZORA-def2-SVP, ZORA	+53.5	-53.2	antiferromagnetic
B3LYP/def2-TZVP, non-rel.	+78.7	-78.1	antiferromagnetic
B3LYP/DKH-def2-TZVP, DKH	+127.84	-126.87	antiferromagnetic
B3LYP/ZORA-def2-TZVP, ZORA	+127.84	-126.87	antiferromagnetic
TPSSh/def2-SVP, non-rel. (10% HF exchange)	-174.9	+173.4	ferromagnetic
B3LYP/def2-SVP, non-rel. (20% HF exchange)	-38.0	+37.8	ferromagnetic
PBE0/def2-SVP, non-rel. (25% HF exchange)	+30.9	-30.8	antiferromagnetic

Basis set: the sign change is an SVP artefact. With TZVP-level basis sets all three calculations give antiferromagnetic coupling: $J(3) = -78.1 \text{ cm}^{-1}$ non-relativistic, -126.87 cm^{-1} with DKH and -126.87 cm^{-1} with ZORA. DKH and ZORA now agree to 0.003 cm^{-1} in $E(\text{HS}) - E(\text{BS})$ (they differed by 32 cm^{-1} with SVP), and the non-relativistic value changes sign on its own when the basis is enlarged. The ferromagnetic SVP result and the DKH/ZORA disagreement therefore come from the small SVP basis, not from relativity. With the larger basis the relativistic correction is clean: scalar relativity shifts J by -49 cm^{-1} , i.e. it strengthens the antiferromagnetic coupling by about 62 %.

Functional: the largest uncertainty. With the same SVP basis $J(3)$ is $+173 \text{ cm}^{-1}$ for TPSSh (10% exact exchange), $+38 \text{ cm}^{-1}$ for B3LYP (20%) and -31 cm^{-1} for PBE0 (25%), a monotonic trend with the amount of exact exchange spanning 204 cm^{-1} . This is the well-known functional dependence of broken-symmetry DFT and is larger than any of the relativistic shifts.

How important are the relativistic effects on J?

The relativistic Hamiltonian changes the total energy by tens of hartree, but almost all of that is the core-electron stabilisation, which is the same in the high-spin and the broken-symmetry state and cancels in J. What remains for J is a correction of about 49 cm^{-1} ($\approx 62\%$ of $|J|$): for a 3d metal like Cu this is small in absolute terms but not negligible, because the exchange coupling itself is a small energy. DKH and ZORA give the same answer once the basis set is adequate, so for J the choice between the two scalar-relativistic Hamiltonians does not matter. The comparison also shows that such small energy differences need a converged basis set before a relativistic effect can be read from them — the SVP basis prescribed for the exercise is too small for that — and that the exchange-correlation functional, not relativity, is the main source of uncertainty in the computed J.